scientific reports

Check for updates

OPEN

Neurovascular and inflammatory effects of biperiden in the acute phase of moderate traumatic brain injury: evidence from a non-human primate model

Viviam Sanabria[1]✉, Christiane Gimenes[1], Simone Romariz[1], Amarildo Souza Gois[2], Matheus B. Braga[1], Michele Longoni Calió[2,3], Juliana Lima Willers[2], Maira L. Foresti[2,3], Luiz Eugênio Mello[2,3] & Beatriz M. Longo[1]✉

Traumatic brain injury (TBI) leads to extensive structural damage, neuronal loss, and functional deficits, along with marked disruption of cholinergic signaling and acute surges in acetylcholine. Biperiden, a muscarinic cholinergic antagonist, has emerged as a potential neuroprotective agent due to its ability to modulate synaptic plasticity and reduce excitotoxicity. Here, we evaluated the acute effects of biperiden in a translational non-human primate model of moderate TBI. Marmosets (*Callithrix jacchus*, *n* = 39) subjected to lateral fluid percussion injury (LFPI) received intraperitoneal biperiden (8 mg/kg) beginning 6 h post-injury, followed by two additional doses administered at 8 h intervals. At 24 h post-injury, brain tissue and serum were assessed using histology, immunofluorescence, and Single Molecule Array (SIMOA). Twenty-four hours post-trauma, biperiden treatment reduced hippocampal neuronal degeneration, decreased UCH-L1 levels, and attenuated astrocyte activation compared to saline-treated controls. Although biperiden is a clinically characterized muscarinic antagonist, the direct muscarinic target was not assessed in this primate model. These findings provide preliminary evidence that repeated dosing with biperiden may modulate early neurovascular and glial responses and provide acute neuroprotection following moderate TBI in marmosets, mitigating neuronal injury, excitotoxicity, and inflammation. This work highlights biperiden as a promising therapeutic candidate for early intervention after TBI.

Keywords Acute, Glia, Vessels, Biperiden, Hematomas, SIMOA

Traumatic brain injury (TBI) occurs when an external force impacts the head, potentially leading to anatomical lesions, brain dysfunction, and neuronal damage[1]. Following TBI of severity (moderate and severe), the risk of developing neurodegenerative disorders such as dementia, Parkinson's, Alzheimer's disease, and the risk of epilepsy increases[2–4]. Therefore, it represents a significant cause of long-term disability, socioeconomic burden, and mortality[5,6].

To investigate the mechanisms underlying TBI and potential neuroprotective therapies, animal models have been developed to reproduce distinct injury paradigms and to elucidate the primary and secondary sequelae associated with human head trauma[7]. One of these models is the lateral fluid percussion injury (LFPI), which reproduces pathologies associated with human TBI, including neuronal loss, inflammation, gliosis, vascular disruptions, hemorrhages, and molecular disturbances[8,9]. Although non-invasive models (e.g. CHIMERA) better mimic closed-skull injury[10], LFPI remains widely validated and was selected for its ability to deliver a controlled, reproducible insult producing moderate focal and diffuse damage[11], suitable for studying early neurovascular, inflammatory, and neuronal responses.

[1]Laboratory of Neurophysiology, Department of Physiology, Universidade Federal de São Paulo, São Paulo, Brazil. [2]Laboratory of Neurobiology, Department of Physiology, Universidade Federal de São Paulo, São Paulo, Brazil. [3]Instituto D'Or de Pesquisa e Ensino, São Paulo, São Paulo, Brazil. ✉email: calvo.sofia10@unifesp.br; biamonteironeuro@gmail.com

In this research path, rodents (mice and rats) are the most widely used species[12]. However, it's important to note that rodents per se are not the best models to replicate the biomechanical and physiological parameters of human TBI[12], mainly because their smooth or lissencephalic brains lack the cortical folding (gyri and sulci) present in the human gyrencephalic brain[13]. This is a significant limitation in TBI research, as the structural difference influences how the brain deforms within the skull[14].

In lissencephalic brains, trauma is uniformly distributed near the surface. Conversely, sulci cause concentrated stress at their bases, resulting in maximum mechanical stress occurring deeper in gyrencephalic brain tissue[13]. Moreover, rodents have less white matter than marmosets, sheep, and pigs, a distinction that is particularly relevant in studies of diffuse axonal injury or edema[13]. The gray-to-white matter ratio in the marmoset brain closely resembles that of humans[13].

The marmoset (*Callithrix jacchus*), a New World primate that diverged from humans approximately 45 million years ago, has a quasi-gyrencephalic brain with underdeveloped sulci[15]. This brain architecture, combined with the marmoset's primate-specific neurobiology, higher cognitive complexity, and its closer translational relevance[16], positions it as a superior model for studying the mechanisms and consequences of TBI. Even though ferrets[17] and pigs[18] have also been investigated and both feature gyrencephalic brains, marmosets provide a better understanding of human neurobiology, offering hope for more effective treatments and interventions (e.g., potential therapeutic development, biomarkers) that may require primate-specific responses and offer increased regulatory and clinical relevance[19].

TBI profoundly disrupts cholinergic signaling[20], with hippocampal acetylcholine levels increasing by approximately 74% immediately post-injury, and similar alterations have been reported after fluid percussion injury (a widely used TBI model)[21–23]. Biperiden, a muscarinic cholinergic antagonist, has been proposed as a potential modulator of neural plasticity following TBI, where it delays seizure onset and reduces severity[24]. Studies deploying biperiden to alter disease progression (epileptogenesis) after TBI in humans have, so far, not clearly shown its effectiveness[25]. There is a great debate as to whether models of status epilepticus (SE), the condition in which biperiden has mostly been tested, are the best strategy to study TBI[23].

In the present study, our aim is to investigate the effects of biperiden during the acute phase (24 h) of moderate TBI in a non-human primate model, to observe and analyze structural and cellular alterations within the brain. We hypothesize that biperiden treatment during the acute phase of moderate TBI may affect neuronal and glial injury. We hope to shed light on relevant mechanistic understanding in ongoing clinical trials with biperiden in TBI[26].

Methods

Animals

Thirty-nine adult marmosets (*C. jacchus)* (24 males, 15 females; age: 2–6 years; weight: 180–380 g) were obtained from the Centro de Manejo e Conservação de Animais (CeMaCAS) and transported to the Federal University of São Paulo primate facility. They were housed individually in wire cages (50 cm × 50 cm × 50 cm) under controlled conditions, with stable room temperatures (25 ± 2 °C) and a 12-hour light-dark cycle starting at 7:00 AM. Animals received enrichment items (branches and wooden swings), free access to water, and fresh fruit twice daily. All procedures followed ARRIVE, NIH, and the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA) guidelines, with approval of the Board for Ethics in the Use of Animals (CEUA, Comissão de Ética no Uso de Animais), an institutional ethics committee of the UNIFESP, protocol nº 5,138,271,222, and were conducted following the guidelines for animal care and use of laboratory animals. All efforts were made to minimize the number of animals and their suffering.

Experimental design and groups

Animals were randomly assigned to four groups. The investigators were blinded to group allocation throughout the experiments and during outcome assessments. The experimental design is shown in Fig. 1a.

- **NAÏVE**: Intact animals that underwent neither craniectomy nor LFPI. $n = 8$ (5 females and 3 males).
- **SHAM-SAL**: Animals underwent craniectomy but did not receive the LFPI impact. They were treated with a saline solution injection via the intraperitoneal (i.p.) route. $n = 9$ (4 females and 5 males).
- **TBI-SAL**: Animals underwent LFPI and were treated with saline solution injections (i.p.). $n = 9$ (3 females and 6 males).
- **TBI-BIP**: Animals underwent LFPI and were treated with biperiden injections (i.p). $n = 8$ (2 females and 6 males).

LFPI model of brain trauma

Animals were anesthetized with isoflurane (3% for inhalation and 2% for maintenance, at a flow rate of 1–1.5 L/min in freely breathing oxygen), followed by an intradermal injection of lidocaine (10 mg/kg) for local anesthesia. A 15 mm midline incision was made to expose the skull, and a 5 mm burr hole was drilled using an electric drill (Stoelting, Wood Dale, IL), ensuring the integrity of the dura mater in all trauma regions. To minimize thermal and mechanical damage, the procedure incorporated continuous irrigation with cold saline, a low drill speed, and intermittent drilling. At the craniectomy site, a cannula (a female Luer lock) was fixed with dental cement. After the cannula was securely placed and filled with sterile 0.9% saline solution. The female Luer lock on the animal's skull was then connected to the male Luer lock of the fluid percussion device (Model FP301 Signal Conditioner, AmScien Instruments).

Stereotaxic coordinates for trauma induction were: +1.78 mm anterior-posterior (AP) to bregma, +10 mm medial-lateral (ML), + 13 mm dorso-ventral (DV). The LFPI model was applied at a 17° angle, with a brief (10–15 ms) transient fluid-pressure pulse impacting the exposed dura. Pulse pressures were measured by an extracranial

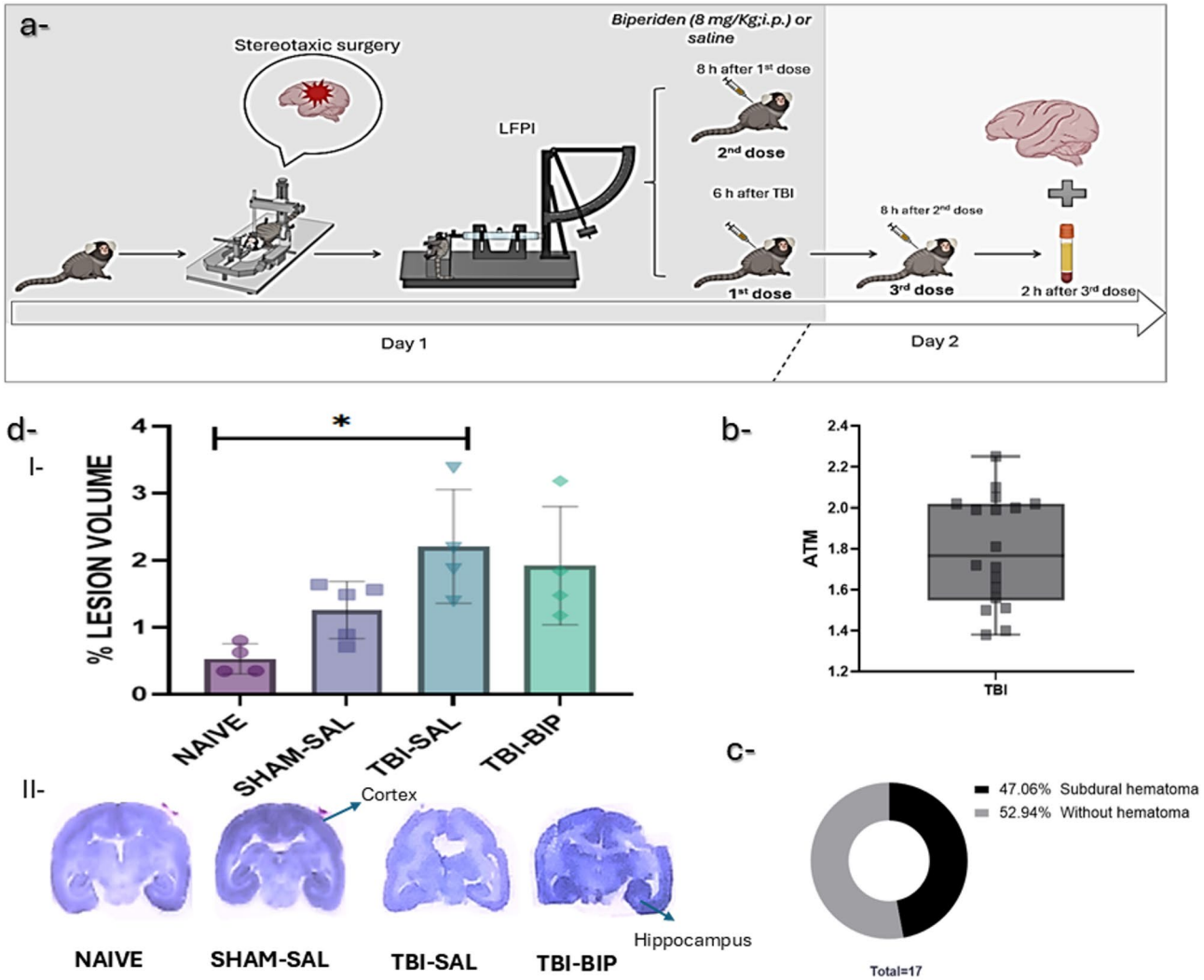


Fig. 1. Experimental design and LFPI model standardization. **a**- Overview of the experimental design. Stereotaxic surgery and induction of traumatic brain injury (TBI) by lateral fluid percussion injury (LFPI) were performed on day 1 in *C. jacchus*. Twenty-four hours after trauma, animals were euthanized and perfused for histological analyses. **b**-General pressure value in atmospheres (ATM). **c**-Percentage of Hematoma's presence in animals submitted to TBI ($n = 17$). **d**-***I***. Percentage of lesion volume among the groups and ***II.*** Images of Nissl staining across the groups; arrows indicate the location of the cortex and hippocampus. Statistical significance: *$p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$. Data are shown as the mean ± standard deviation of the mean.

transducer and recorded on a storage oscilloscope (Fig. 1a). After the LFPI, the cannula was removed from the skull, and the animal was sutured with 2 – 0 non-absorbable nylon.

During stereotaxic surgery, animals were maintained normothermic (36–38 °C) using a thermostatically controlled heating blanket, with rectal temperature continuously monitored (RightTemp Jr., Kent Scientific, USA). Physiological parameters, including heart rate, respiratory rate, and oxygen saturation, were continuously monitored using a pulse oximeter with paw sensors (Kent Scientific). Mucous membrane color was also assessed to ensure stable physiological conditions. At the time of injury induction, animals were fully recovered from isoflurane anesthesia. Animals received post-surgical care, including pentabiotic (0.1 mL/kg, i.m.) and flunixin meglumine (1 mg/kg, i.p.). They were kept on a heated blanket and hydrated until they had fully recovered. At 24-hour post-trauma, animals were deeply anesthetized with pentobarbital (100 mg/kg, i.p.) and transcardially perfused with phosphate-buffered saline, followed by 4% paraformaldehyde.

Biperiden treatment

Animals were treated with biperiden hydrochloride dissolved in 0.9% saline at a concentration of 8 mg/Kg, i.p. (Cristália, Brazil). Saline (0.9%) served as a vehicle. Treatment started six hours after the induction of TBI or six hours after the induction of anesthesia in the sham-operated marmosets. The second dose was administered 8 h after the first, and the third dose was administered 8 h after the second. The biperiden dose of 8 mg/Kg

was selected based on prior rodent studies demonstrating its safety and efficacy in modulating excitotoxicity and epileptogenesis[24,27]. The dose-dependent effects of similar anticholinergic treatments were demonstrated by Benassi et al.[27]; therefore, it can be safely administered after TBI.

Brain tissue preparation

Twenty-four hours after LFPI-TBI, the animals were deeply anesthetized with pentobarbital (100 mg/kg, i.p.). A sternotomy was made to access the heart. Then, the animals were perfused through the heart. Perfusion started with 100 mL of 0.1 M PBS solution (phosphate buffer solution; 5.52 g of monobasic sodium phosphate plus 21.88 g of dibasic sodium phosphate), then 100 mL of 4% paraformaldehyde (diluted in PBS solution).

The brains were removed and post-fixed in 4% formaldehyde, then cryoprotected in 30% sucrose (diluted in PBS). They remained in this solution until they showed signs of dehydration, a procedure that usually occurred within 24 h. Immediately after this period, the brains were dried, frozen on dry ice, embedded in Tissue-Tek, frozen at -80 °C, and cut into a series of 40 µm-thick coronal sections with a cryostat (Leica CM150, Nussloch, Germany). Brain sections were systematically collected into five 6-well plates while preserving rostrocaudal order to ensure unbiased neocortical sampling. Sections were stored in an antifreeze solution (300 g of sucrose, 500 mL of PBS, and 300 mL of ethylene glycol) at -20 °C until processed for histological and immunohistochemical analysis. Brain sections were selected according to the stereotaxic atlas[28], ranging from +3.64 to −1.84 mm in the anterior–posterior axis relative to bregma (dorsal hippocampus), ensuring consistent sampling of both neocortical and hippocampal ipsilateral regions for histological assessment. Based on a pilot study, systematic random sampling was applied: every 6th section within each piece was selected.

Histological nissl staining

Brain slices were mounted on gelatin-coated glass slides. The slides were dehydrated through a graded series of solutions (xylene, 100%, 96%, and 70% ethanol, followed by Milli-Q water). Sections were then stained with 1% cresyl violet (Sigma) prepared in distilled water and filtered for 8 min. This was followed by differentiation in acetic acid in 70% ethanol (250 mL of 70% ethanol with 2.5 mL of glacial acetic acid) for 5 s. Subsequently, the sections were dehydrated in an ascending ethanol series (70%, 96%, and 100%), cleared in xylene for 2 min, and finally mounted with Entellan mounting medium and coverslipped. Slides were imaged (6 brain sections for each animal), and the lesion area was assessed among the groups: NAIVE, SHAM-SAL, TBI-SAL, and TBI-BIP. Images were analyzed and imported into Fiji (http://fiji.sc), a distribution of ImageJ (National Institutes of Health, http://imagej.nih.gov/ij, version 2.14.0). Brain volume was estimated by the Cavalieri method[29]. Sections were scanned with an HP LaserJet Managed MFP E42540 scanner, and a point grid was overlaid on the images for analysis. Volume was calculated as:

$V = t \times a/p \times P$ [29].

where t is the section thickness, a/p is the area per grid point, and P is the total number of points hitting the tissue[29]. The sectional area was estimated using a point-counting grid.

The percentage of lesion volume was calculated as per the following formula[30]:

$$Lesion\ volume = \frac{volume\ of\ contralateral\ hemisphere - volume\ of\ ipsilateral\ hemisphere}{volume\ \ contralateral\ hemisphere} * 100$$

Fluoro-Jade® B

Fluoro-Jade® B (FJ-B, AG310 Merck-Millipore) was used to stain degenerating neurons. Fixed sections were mounted on glass slides, and FJ-B staining was performed as follows: The slides were mounted and let dry at room temperature for 3 to 5 days. Then, the slices were placed in an oven at 37 °C for 20 min.

The slices were then immersed in 100% ethanol for 3 min, followed by 1 min each in 70% ethanol and distilled water. The slides were then incubated in 0.06% potassium permanganate ($KMnO_4$) in distilled water for 15 min. Next, the slides were rinsed in distilled water and transferred to a 0.0001% FJ-B staining solution in 0.1% acetic acid for 30 min. The FJ-B working solution was prepared from a stock FJ-B solution (0.01% in distilled water). After rinsing, the slides were coverslipped with DPX.

The slides were analyzed using a 20X objective and a computer-based digitizing image system (Zeiss Axiovert; Carl Zeiss, Germany) connected to an Axiocam 208 color camera. Image analysis was conducted in ImageJ using the Analyze Particles plugin for cell counting in areas of interest. Because the lesion site was often absent or too damaged for reliable analysis (Fig. 1), we quantified only the adjacent intact tissue—the neocortex (motor, parietal, and temporal cortices) and hippocampus (CA1, CA3, and dentate gyrus (DG))—located within 2 mm of the lesion border.

Immunofluorescence

Immunofluorescence was performed to identify astrocytes, microglia, and vessels. Free-floating sections (six brain slices per animal) were washed five times with PBS solution to remove all anti-freezing solution and then incubated in a blocking solution (250 µL of Triton X-100, 90 mL of 0.1 M PBS, and 40 µL of goat serum) for 30 min at room temperature under constant stirring. After blocking, cells were incubated overnight with the following primary antibodies: polyclonal rabbit anti-Iba-1 (1:1000, Wako, 019-19741) for microglia/macrophages, and monoclonal mouse anti-glial fibrillary acidic protein (GFAP) (1:1000, Sigma, G3893) for astrocytes, diluted in blocking solution.

The following day, after washing the sections in PBS, they were incubated for 2 hours with secondary antibodies (anti-mouse Alexa Fluor 488, anti-rabbit Alexa Fluor 568, 1:600, all from Invitrogen) in blocking solution. For vessel staining, after three PBS washes, the sections were incubated with Tomato lectin (TL; Lycopersicon esculentum; 1:200; Vector Laboratories; FL-1171-1) for 1 hour. Nuclei were stained with DAPI

(4',6-diamidino-2-phenylindole, 1:10,000; Thermo Fisher Scientific, Carlsbad, CA, USA). Finally, the sections were coverslipped with Fluoromount G™ (Thermo Fisher Scientific).

For the analysis of Iba-1, GFAP, and TL, the slides were examined using a computer-based digitizing image system (Zeiss Axiocam 506 color; https://www.zeiss.com/axiocam) with a 20X objective. The laser and detector were maintained at constant settings for both cases during the acquisition of each staining set.

For each slide, cells were qualitatively evaluated for their localization and morphology within ipsilateral brain regions affected by trauma, including the neocortex (motor, parietal, and temporal cortices) and hippocampus (CA1, CA3, and DG). Slides and resultant images were coded, and fluorescence signals were quantified. Stained cells were quantified by branch count and average branch length using skeleton and vascular density analysis in Fiji (http://fiji.sc), a distribution of ImageJ software (National Institutes of Health http://imagej.nih.gov/ij), version 2.14.0. Vessel count was also performed manually.

Image analysis was conducted using Fiji (ImageJ, National Institutes of Health, version 2.14.0) using 2D projections. From the six brain sections obtained per animal, three were randomly selected by an observer blinded to the experimental groups. Within each region, the neocortex (motor, parietal, and temporal cortices) and the hippocampus (CA1, CA3, and DG) were divided into three regions of interest (ROIs) per subregion, yielding a total of 18 ROIs per section. Images were acquired for each ROI, and approximately 100 cells were analyzed per region, ensuring consistent sampling density across animals and brain areas.

For Iba-1 and GFAP analyses, the skeleton-derived parameters—number of branches and average branch length—were subjected to agglomerative hierarchical clustering using the average linkage method. This approach groups data based on their similarity, producing a hierarchical dendrogram in which the branching levels represent the degree of dissimilarity between clusters. In the dendrogram, clusters are depicted as nodes, and the vertical axis represents the degree of similarity between them. Naive groups are represented by numbers from 1 to 4, SHAM-SAL from 5 to 8, TBI-SAL from 9 to 12, and TBI-BIP from 13 to 16.

Single molecule array (SIMOA)®

Detection of blood-derived brain biomarkers associated with TBI was performed using the SIMOA platform (Quanterix Corporation, Lexington, MA). Total tau (T-tau; a marker of neuronal function), Neurofilament Light Chain (NfL, an indicator of neuronal damage), Glial Fibrillary Acidic Protein (GFAP; a marker of astrocytic damage), and Ubiquitin C-Terminal Hydrolase L1 (UCH-L1; an early marker of neuronal injury). These biomarkers were selected for their sensitivity to TBI-related damage. Serum biomarker levels were quantified using the Neurology 4-Plex B (N4PB) assay, a specific immunoassay kit that detects NFL, total tau, GFAP, and UCHL-1. The samples were thawed, vortexed, and centrifuged for the analysis. The analysis was performed at Richet/IDOR in Rio de Janeiro using the SIMOA HD-1 device.

Serum samples from animal models were thawed, and 60 µL of each sample was pipetted into each well of a 96-well plate, along with the assay-validated calibrators. The device was loaded with the necessary reagents and supplies, including pipettes, commercial kit beads, discs, and the sample plate. After adjusting the parameters, the assay was initiated and lasted approximately 4–5 h. The automated calibration cycles, pipetting, washing, incubation, matrix transfer, and signal reading recorded the intensity of the fluorescent signals emitted by the microparticles, expressed in picograms per milliliter (pg/mL). The calibration curve converted the signals obtained from the samples into quantifiable concentrations of the biomarkers, a conversion typically performed automatically by the SIMOA software.

A total of six samples were excluded from the final analysis due to pre-analytical coagulation, which compromised sample integrity and prevented reliable quantification.

Statistical analysis

Sample size calculation was performed a priori using G*Power (v 3.1.9.7). The calculation is based on one-way ANOVA (fixed effects, considering four groups), a small effect size ($f = 0.19$), a significance level of 5% ($\alpha = 0.05$), and a statistical power of 0.95. Effect size was calculated using Cohen´s d, and randomization and group allocation were performed by blind investigators.

Normality and homogeneity of variance were assessed using the Kolmogorov-Smirnov and Levene tests, respectively. Due to the absence of normality and comparing independent groups, the non-parametric Kruskal-Wallis (KW) test was applied appropriately, followed by the Dunn post-hoc test. As an exploratory analysis, hierarchical clustering with the average linkage method and Euclidean distance was employed. Correlation matrix, though the Spearman r was employed to identify immunohistochemical variables. Principal component analysis (PCA) was performed to explain the greatest variation in the immunohistochemical variable and reduce data dimensionality. All variables were standardized (mean-centered and scaled to unit variance) prior to analysis.

Data are presented as the mean ± standard deviation (SD), and P-values < 0.05 are considered statistically significant. Data analyses were performed using GraphPad Prism version 11.0 (GraphPad Software Inc., San Diego, CA., http://www.graphpad.com), R version 4.3.2, and RStudio.

Results

Moderate LFP-induced TBI in marmosets resulted in variable hematoma formation

Following parameter standardization for inducing moderate TBI in marmosets, the optimal pressure range was determined to be 1.38–2.25 atm (mean 1.78 ± 0.26 atm; Fig. 1b), with a 17° impact angle consistently producing moderate injury, as established in our previous study (Sanabria et al., 2026)[31].

Of the 39 animals, 2 died during anesthesia, and 3 died due to trauma. The acute mortality rate in the TBI group (within 24 h post-impact) was 15% (3/20), with all deaths occurring immediately after injury. Among the 17 animals that survived LFPI, 8 (47%) developed subdural hematomas and 2 (12%) developed epidural

hematomas (Fig. 1c). Hematomas were identified at euthanasia by blind investigators after removal of the cranium and dura mater.

Among animals with hematomas ($n = 8$), 5 belonged to the TBI-SAL group (1.66–2.25 atm) and 3 to the TBI-BIP group (1.50–1.61 atm), with a mean pressure of 1.81 ± 0.07 atm. Animals without hematomas were exposed to pressures ranging from 1.38 to 2.02 atm.

Modest effect of biperiden on lesion volume

The percentage of lesion volume induced by TBI was quantified using Nissl staining. Lesion volume was calculated by subtracting the volume of the ipsilateral hemisphere from that of the contralateral hemisphere. Statistical analysis with the Kruskal–Wallis test (KW (3) = 10.13; $p = 0.0051$) revealed significant differences between the Naïve and TBI group. Specifically, TBI-SAL animals displayed a significantly extended lesion volume compared with Naïve animals ($p = 0.016$; Fig. 1d).

Neuronal degeneration was significantly higher in the TBI-SAL group in CA1

FJ-B staining was employed to evaluate neuronal degeneration in the neocortex (motor, parietal, and temporal cortices) and hippocampus (CA1, CA3, and DG). In the neocortex (across the three analyzed areas), a statistically significant group difference was revealed (KW(3) = 10.42; $p = 0.0038$). Post-hoc analysis (Dunn's test) indicated that both TBI-SAL (145 ± 43.01; $p = 0.024$) and TBI-BIP (132 ± 39.37; $p = 0.047$) groups exhibited significantly greater neuronal degeneration compared with the Naïve group (20.75 ± 16.30; Fig. 2b I). When we analyzed the cortical regions independently, statistically significant differences were observed in the parietal (KW(3) = 11.35; $p = 0.0014$) and temporal cortex (KW(3) = 11.18; $p = 0.0016$). In both cortical regions, TBI groups (TBI-SAL and TBI-BIP) had more neuronal degeneration compared with the Naive group (See Supplemental Materials and Figure S1).

In the hippocampus, the CA1 region showed statistically significant group differences (KW(3) = 10.94; $p = 0.0022$). TBI-SAL group (126 ± 37.73; $p = 0.007$) displayed significantly higher neuronal degeneration compared with the Naïve group (21.25 ± 8.92; Fig. 2b II). While in the CA3 region, no statistically significant differences were detected among groups. In the DG, statistically significant differences were observed (KW(3) = 11.02; $p = 0.0020$). The SHAM-SAL group (84.60 ± 22.66; $p = 0.043$) exhibited significantly higher neuronal degeneration compared with the TBI-BIP group (30.25 ± 7.84; Figs. 2b III-IV).

Biperiden increased vessel numbers across cortical and hippocampal regions, whereas vascular density remained unchanged

Concerning the number of vessels in the different areas analyzed. In the neocortex (across the three analyzed areas), we observed statistically significant differences among the groups (KW(3) = 10.38; $p = 0.0040$). The TBI-SAL group had an increase in the number of vessels (62.08 ± 19.28) compared to the Naïve group (30.21 ± 8.22; $p = 0.0250$; Fig. 3b I). However, when we analyzed different parts of the neocortex independently, we observed statistically significant differences in the parietal cortex (KW(3) = 9.50; $p = 0.0093$) and in the temporal cortex (KW(3) = 11.47; $p = 0.0012$). In the parietal cortex, the SHAM-SAL (59.46 ± 20.81; $p = 0.009$) group and the TBI-BIP (54.00 ± 3.65; $p = 0.017$) group had more vessels than the Naive (31.13 ± 8.04; see Supplemental Material and Figure S2).

In the hippocampus, in the CA1 region, we observed statistically significant differences among the groups (KW(3) = 12.51; *$p = 0.0002$).* The SHAM-SAL had more vessels (57.71 ± 12.59) compared to the Naïve (22.56 ± 2.78; $p = 0.011$) and the TBI-SAL (30.71 ± 8.00; $p = 0.04$). Moreover, the TBI-BIP group also had more vessels (57.29 ± 5.99) than the Naïve ($p = 0.012$) and more than TBI-SAL ($p = 0.039$; Fig. 3b II).

Similarly to the cortex, in CA3 (KW(3) = 11.15; $p = 0.0017$) and DG (KW(3) = 9.42; $p = 0.0087$) we observed statistically significant differences among the groups, in both cases the SHAM-SAL [in CA3:59.29 ± 18.17, in DG:67.39 ± 24.75] group had more vessels than the Naïve group [in CA3:24.90 ± 3.76, in DG:24.67 ± 6.68; Figs. 3b III-IV].

No significant differences in vascular density were observed among the groups across the regions analyzed (Figs. 3c I-VIII).

Astrocytes in the TBI-SAL group exhibited reduced astrocytic branching, while TBI-BIP had longer processes

A skeleton analysis was performed to assess astrocyte morphology by measuring branch number and length in the neocortex and hippocampus.

Significant differences in astrocyte branch number were observed only in the cortex (KW(3) = 14.49; $p < 0.0001$). Post hoc analysis showed that both TBI-SAL (61.25 ± 11.00 branches; $p = 0.047$) and TBI-BIP (47.50 ± 10.41 branches; $p = 0.0024$) had significantly fewer branches compared to the SHAM-SAL group (326.2 ± 97.61 branches; Fig. 4b I-IV). When cortical regions were analyzed separately, branch numbers also differed significantly across groups. In the motor cortex (KW(3) = 10.94; $p = 0.0003$), SHAM-SAL showed more branches than TBI-BIP (46 ± 6.89; $p = 0.0041$). In the parietal and temporal cortex (both KW(3) = 13.06; $p < 0.001$), both TBI-BIP (Parietal: 48 ± 9.76; Temporal:40.75 ± 28.32) groups exhibited reduced branching compared to SHAM-SAL (Parietal:319 ± 97; $p = 0.0031$; Temporal:342 ± 88 ; $p = 0.007$).

Astrocyte branch length differed significantly between groups in the cortex (KW(3) = 14.11; $p < 0.0001$), with the TBI-BIP group (29.82 ± 9.25 µm) showing longer branches than the TBI-SAL group (3.53 ± 2.27 µm; $p = 0.0097$). Similar differences were observed across all cortical regions—motor (KW(3) = 8.82; $p = 0.01$), parietal (KW(3) = 12.95; $p < 0.0001$), and temporal cortex (KW(3) = 11.76; $p < 0.001$)—with TBI-BIP consistently showing greater branch length than TBI-SAL (see Supplemental Material and Figure S3).

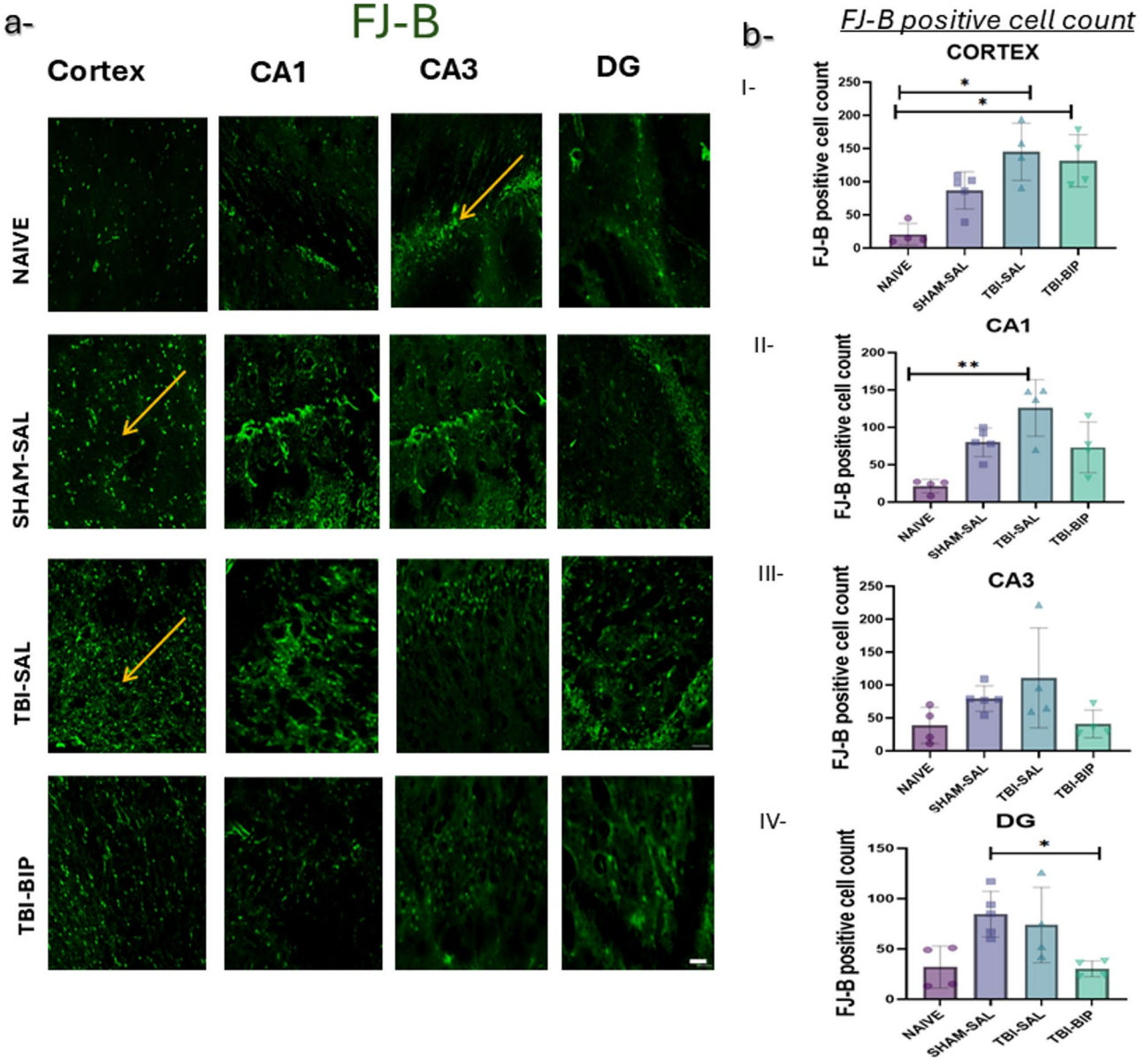


Fig. 2. Fluorojade B (FJ-B) positive cells in neocortex and dorsal hippocampus (CA1, CA3, and DG). **a**- Photomicrographs of FJ-B in the cortex for each group: NAIVE, SHAM-SAL, TBI-SAL, and TBI-BIP. **b**- Graphic representation of the number of FJ-B positive cells in **I**- Cortex, **II**- CA1, **III**-CA3, **IV**-DG. Yellow arrows represent the degenerating neurons. Statistical significance: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$. Data are shown as the mean ± standard deviation of the mean. Scale bars = 40 µm.

In the hippocampus, branch length also differed significantly among groups. In CA1 (KW(3) = 10.47; $p = 0.0036$), CA3 (KW(3) = 12.31; $p = 0.0004$), and DG (KW(3) = 10.94; $p = 0.0022$), the TBI-BIP group showed significantly longer astrocyte branches than the TBI-SAL group. In the DG, the Naive group also exhibited longer branches than TBI-SAL (Figs. 4c III-IV).

Hierarchical cluster analysis based on branch number and average branch length revealed two main clusters for branch number across the cortex and hippocampus, separating SHAM-SAL from the other experimental groups. For branch length, clustering separated TBI-BIP from the other groups in the cortex and CA3, while in the DG, Naive clustered with TBI-BIP, and in CA1, these groups formed separate clusters.

Microglia in the TBI-SAL group exhibited a reduced number of microglia branching, accompanied by longer process length

Skeleton analysis was also performed to evaluate microglia morphology by measuring branch number and branch length in the neocortex and hippocampus (CA1, CA3, DG) (Fig. 5a-b). Regarding the number of microglial branches in the cortex, statistically significant differences were observed among groups (KW (3) = 10.16; $p = 0.0048$). Dunn's post-hoc analysis indicated that the NAIVE (51.75 ± 22.75 branches; $p = 0.044$) group had

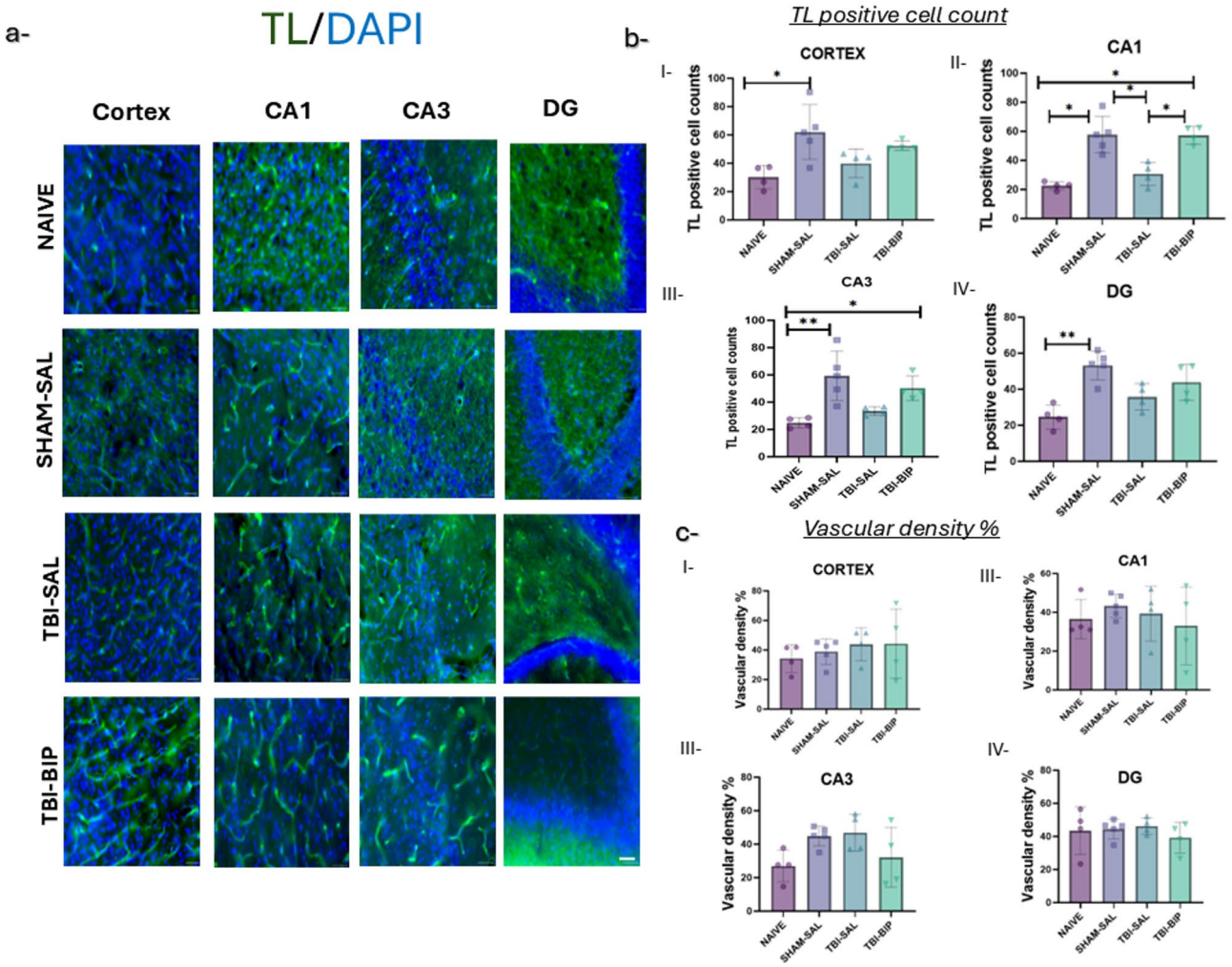


Fig. 3. Tomato Lectin (TL) results. **a**- Photomicrographs of TL immunofluorescence of all groups; cortex and dorsal hippocampus regions (CA1, CA3, and DG). TL (green), DAPI (blue). **b**- TL positive cells count in the **I**-Cortex, **II**-CA1, **III**-CA3, and **IV**-DG. **c**- Percentage of vascular density. **I** -Cortex; **II**- CA1; **III**-CA3; I**V**-DG. Statistical significance: *$p < 0.05$; **,$p < 0.01$; ***,$p < 0.001$; ****,$p < 0.0001$. Data are shown as the mean ± standard deviation of the mean. Scale bars = 40 µm.

a higher number of microglia branches than TBI-SAL (12.25 ± 5.18 branches). Moreover, when we analyzed different parts of the neocortex independently, we observed statistically significant differences in the NAÏVE and SHAM-SAL groups, which had more branches than TBI-SAL [in motor (KW(3) 6.81; $p = 0.0062$), parietal (KW(3) = 7.42; $p = 0.0045$), and temporal (KW(3) = 8.25; $p = 0.0030$) cortices] (See Supplemental Material, S4).

In the hippocampus, CA1 and CA3 showed statistically significant differences in branch number across groups (CA1: KW (3) = 10.92; $p = 0.0022$; CA3: KW (3) = 11.83; $p = 0.0008$). Dunn's post-hoc analysis indicated that the SHAM-SAL (CA1 60.60 ± 26.43 branches; $p = 0.014$; CA3: 61.60 ± 20.38 branches; $p = 0.042$) group had a higher number of microglia branches than TBI-SAL (CA1 13.50 ± 10.88 branches; CA3 19.00 ± 12.19 branches). In DG, statistically significant differences in the number of branches were observed among the groups KW (3) = 13.48; $p < 0.0001$). Dunn's post-hoc analysis indicated that the SHAM-SAL (65.60 ± 20.18 branches) group had a higher number of microglia branches than TBI-SAL (8.00 ± 5.35 branches; $p = 0.0062$) and TBI-BIP (12.75 ± 1.50 branches; $p = 0.042$).

Regarding the microglial average branch length, no statistically significant differences were observed among groups in the cortex, CA1, or DG. However, when we analyze the different parts of the neocortex, only the temporal cortex (KW (3) 3.87; $p = 0.038$) showed significant differences in the microglial branches between the TBI-BIP group (47.32 ± 10.12 µm) and the SHAM-SAL (18.17 ± 4.45 µm). In the hippocampus, *in* CA3, significant differences were detected (KW (3) = 8.07; $p = 0.029$). The TBI-BIP group displayed longer microglial branches (31.45 ± 7.00 µm) compared to the SHAM-SAL group (15.31 ± 1.71 µm; $p = 0.038$).

Hierarchical clustering of microglial morphology, based on branch number and average branch length, revealed distinct grouping patterns. For the number of branches, two main clusters were identified in the cortex and CA1: one comprising the Naïve and SHAM-SAL groups, and another including the remaining experimental groups. In CA3 and DG, the SHAM-SAL group clustered separately from the others. Regarding average branch

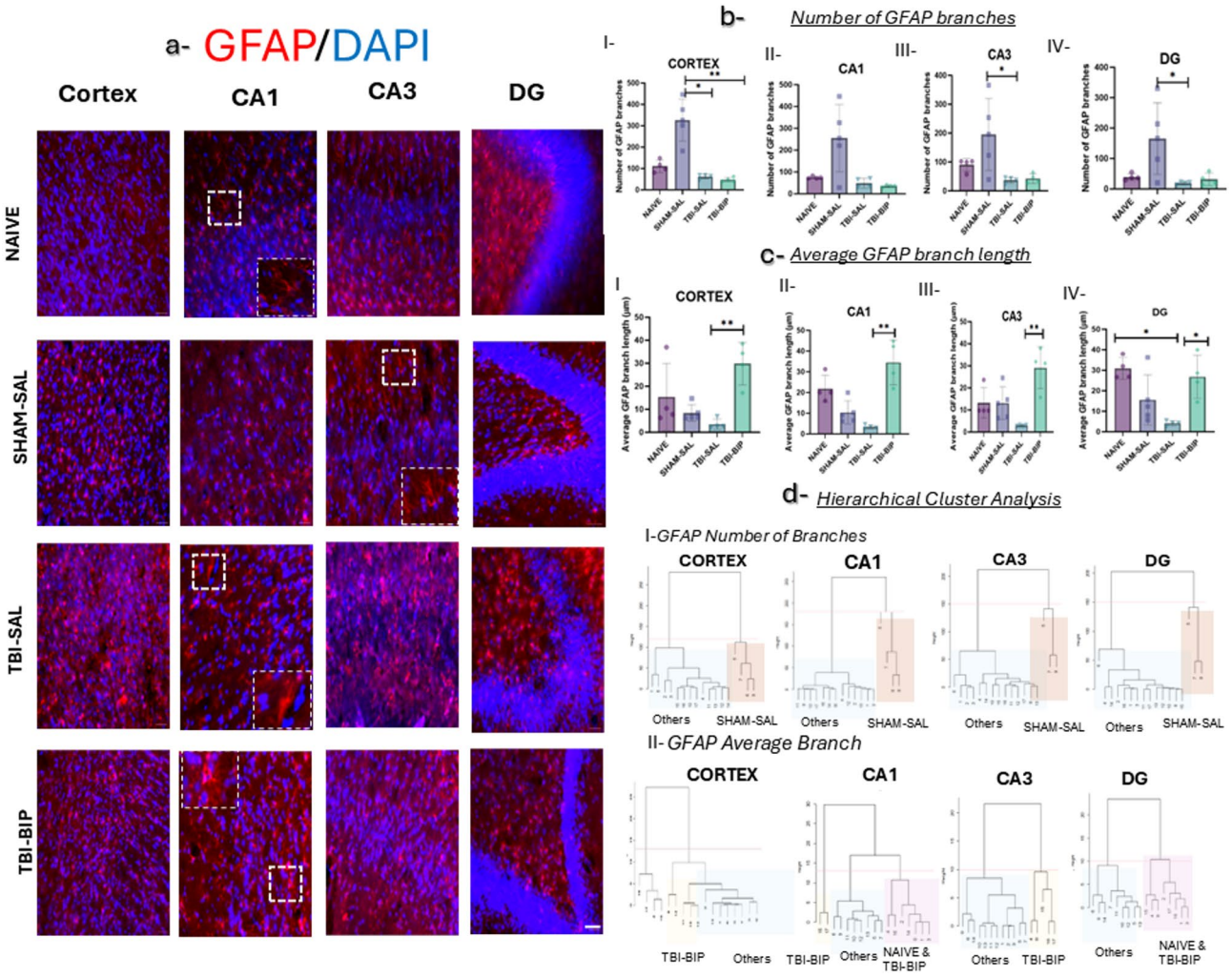


Fig. 4. Astrocytes (GFAP) results. **a**-Photomicrographs of GFAP immunofluorescence of all groups' cortex and dorsal hippocampus regions (CA1, CA3, and DG). GFAP (red), and DAPI (blue). Small boxes with dashed lines indicate the selected zoom-in areas. **b**-Number of GFAP branches **I**-Cortex, **II**-CA1, **III**-CA3, **IV**-DG. **c**-Average GFAP branch length (µm). **I**-CORTEX, **II**-CA1, **III**-CA3, **IV**-DG. **d**-Hierarchical cluster analysis Naive groups are represented by numbers from 1–4, SHAM-SAL from 5–8, TBI-SAL from 9–12, and TBI-BIP from 13–16. **I**- GFAP number branches. Orange boxes represent the SHAM-SAL cluster, and blue boxes represent OTHER (NAIVE, TBI-SAL, and TBI-BIP) cluster. **II**- GFAP Average branch. Orange boxes represent the TBI-BIP or NAIVE and TBI-BIP clusters, and blue boxes represent OTHER (NAIVE, SHAM-SAL, and TBI-SAL) clusters. Statistical significance:*$p < 0.05$; **, $p < 0.01$; ***,$p < 0.001$; ****,$p < 0.0001$. Data are shown as the mean ± standard deviation of the mean. Scale bars = 40 µm.

length, two clusters were observed in the cortex and CA1 (TBI-SAL and TBI-BIP versus the remaining groups), while in CA3, TBI-SAL separated from the others. In the DG, SHAM-SAL and TBI-SAL formed a distinct cluster from the remaining groups.

Correlation matrix and principal component analysis (PCA) among the immunohistochemical variables

In the correlation matrix, we observed a significantly strong positive correlation between the number of GFAP (astrocytes) and Iba-1 (microglia) branches ($r = 0.71$, $p < 0.001$), and a moderate positive correlation between GFAP and Iba-1 average branch length ($r = 0.40$, $p = 0.001$), FJ-B (degenerative neurons) and the number of vessels($r = 0.29$, $p = 0.017$) and the number of vessels and GFAP branch count ($r = 0.26$, $p = 0.029$). In contrast, there was a significant negative correlation between Iba-1 branch count and average branch length (r= -0.36, $p = 0.002$), FJ-B and GFAP branch length (r=-0.41, $p < 0.001$), as well as between GFAP branch count and Iba-1 branch length (r=-0.32, $p = 0.008$; Fig. 6a).

Regarding PCA analysis, we observe that the first two principal components explain 49.02% of the total variance: 30.63% by the first component and 18.39% by the second (Fig. 6b). Therefore, we can say that principal component 1 (PC1) is mainly driven by GFAP and Iba-1 average branch length and FJ-B (Fig. 6c), while PC2 is

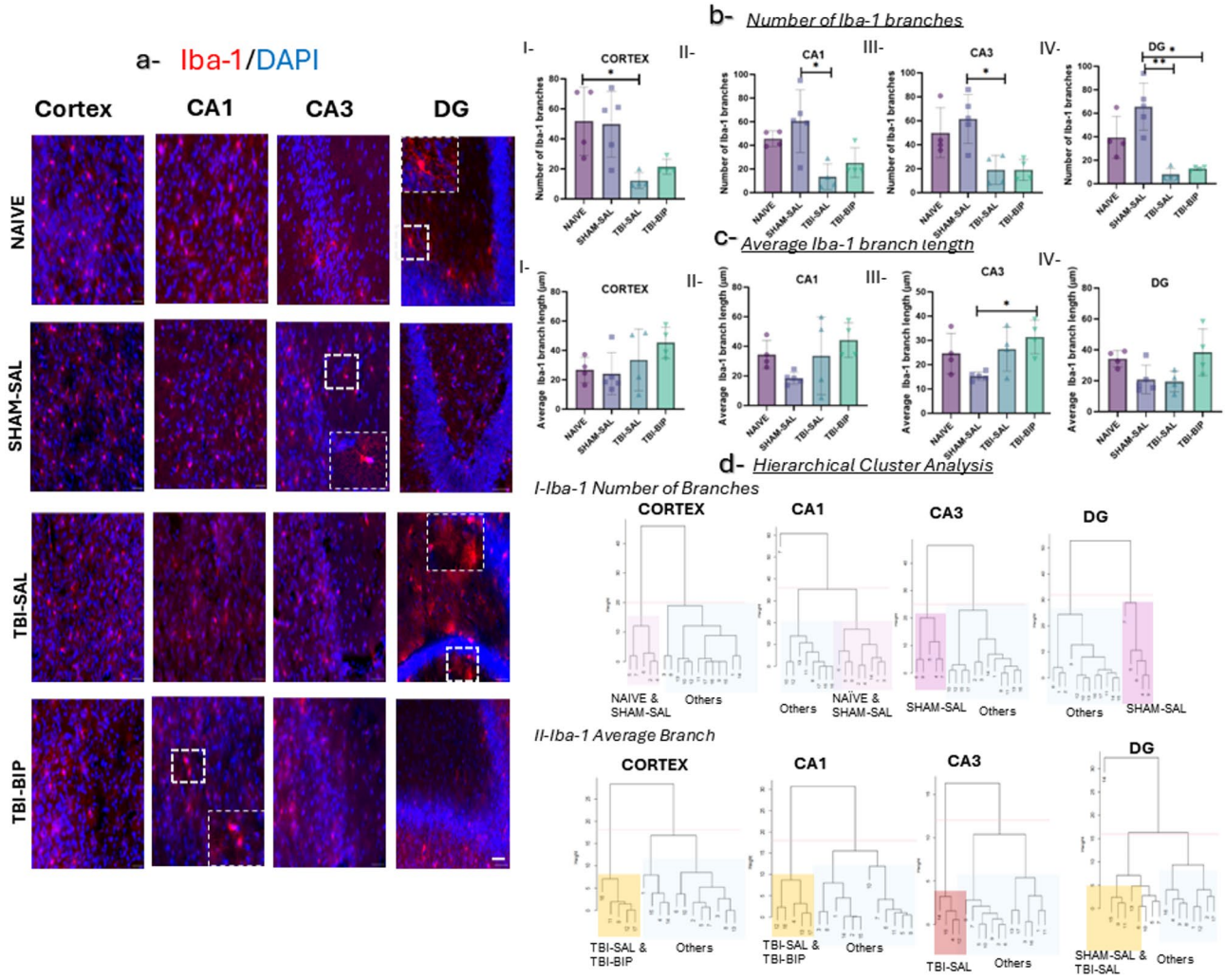


Fig. 5. Microglia (Iba-1) results. **a**-Photomicrographs of Iba-1 immunofluorescence of all group's cortex and dorsal hippocampus regions (CA1, CA3, and DG). Iba-1 (red), and DAPI (blue). Small boxes with dashed lines indicate the selected zoom-in areas. **b**-Number of GFAP branches **I**-Cortex, **II**-CA1, **III**-CA3, **IV**-DG. **c**-Average GFAP branch length (µm). **I**-CORTEX, **II**-CA1, **III**-CA3, **IV**-DG. **d**-Hierarchical cluster analysis Naive groups are represented by numbers from 1–4, SHAM-SAL from 5–8, TBI-SAL from 9–12, and TBI-BIP from 13–16. **I**-Iba-1 number branches. Orange boxes represent the SHAM-SAL cluster, and blue boxes represent OTHER (NAIVE, TBI-SAL, and TBI-BIP) cluster. **II**- Iba-1 Average branch. Orange boxes represent the TBI-BIP or NAIVE and TBI-BIP clusters, and blue boxes represent OTHER (NAIVE, SHAM-SAL, and TBI-SAL) clusters. Statistical significance: $^{*}p < 0.05$; **, $p < 0.01$; ***,$p < 0.001$; ****,$p < 0.0001$. Data are shown as the mean ± standard deviation of the mean. Scale bars = 40 µm.

mainly driven by the number of GFAP and Iba-1 branches, and the number of vessels (Fig. 6d). Two dimensions were selected for the plot shown in Supplemental Material S5.

Early neuronal injury markers were more highly expressed in the TBI-SAL group than in the TBI-BIP group

Regarding the biomarkers of our interest (t-Tau, NfL, GFAP, and UCH-L1). Significant statistical differences were observed only in the UCH-L1 among the groups (KW (2) = 5.79; $p = 0.0457$). Post-hoc test revealed that the TBI-SAL group has higher levels of UCH-L1 (1380 ± 588 pg/mL) compared to SHAM-SAL (211 ± 145 pg/mL; $p = 0.03$) and TBI-BIP (234 ± 77.27pg/mL; $p = 0.04$). No statistically significant difference was observed in the other biomarkers (Figs. 7I-IV).

Discussion

This study investigated the acute (24-hour) effects of biperiden on structural and cellular changes following moderate TBI, as assessed by LFPI, in a non-human primate model (*C. jacchus*). Whereas previous studies have examined the impact of biperiden in SE models[24,27] and LFPI in rodents[32], this is the first study to assess its impact in a quasi-gyrencephalic species, the marmoset. Marmosets offer enhanced translational relevance

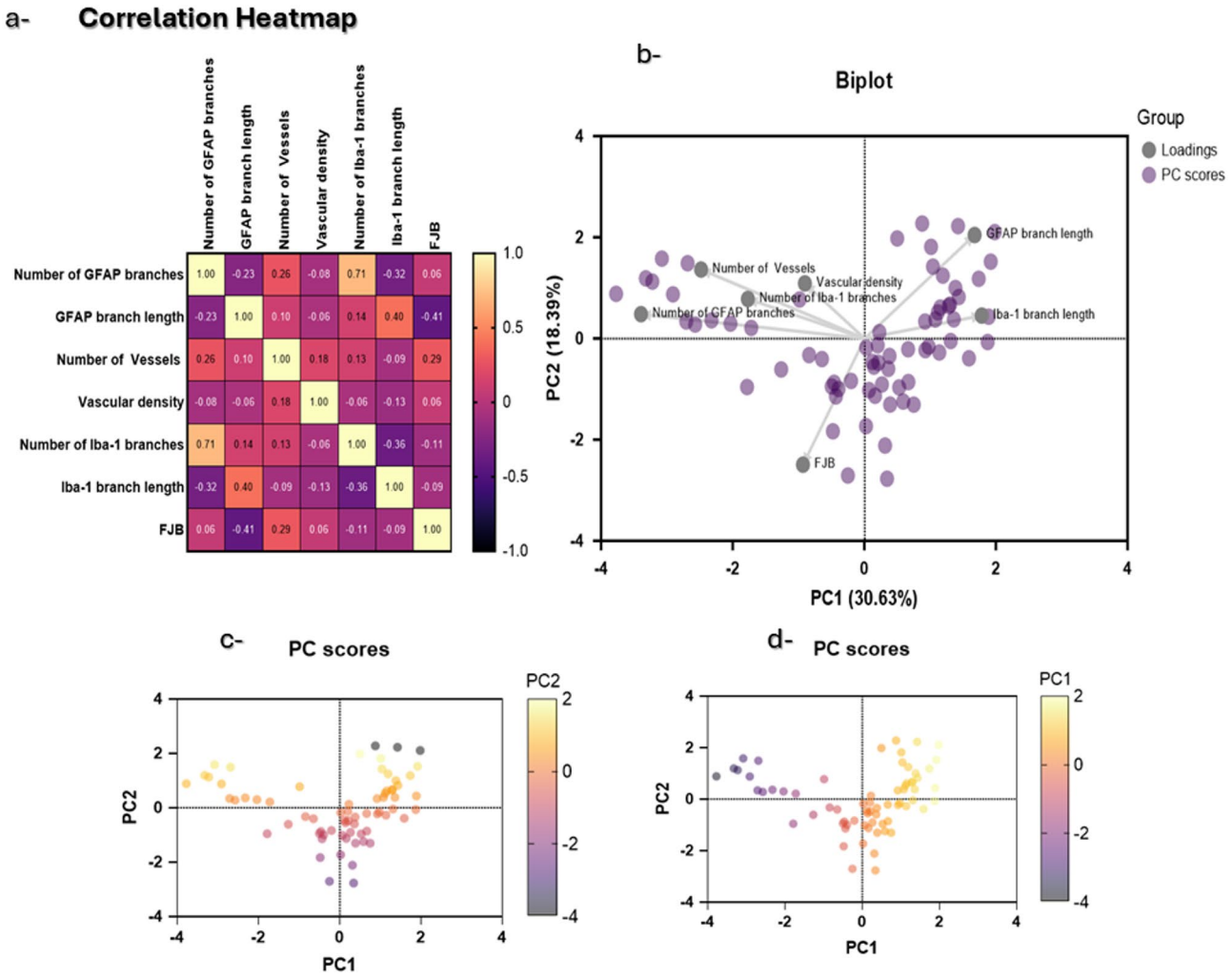


Fig. 6. Correlation matrix and Principal Component Analysis (PCA) of immunohistochemical variables. **a**-Heatmap showing Spearman correlation coefficients for all pairwise comparisons. The scale on the right indicates coefficient values, where 1.0 denotes a perfect positive correlation, − 1.0 a perfect negative correlation, and 0 no correlation. Color intensity ranges from yellow to dark purple: yellow to orange indicates positive correlations, while dark purple indicates negative correlations; stronger relationships are represented by greater color intensity. **b–d** PCA analysis. **b**- PCA biplot showing the first two principal components, with PC1 explaining 30.63% and PC2 explaining 18.39% of the variance. Loadings are shown in gray, and PC scores are shown in lilac. **c**- Distribution of PC scores for PC1. **d**- Distribution of PC scores for PC2. The color scale on the right indicates contribution levels: high contributors in yellow, intermediate in orange, and low in purple.

to human TBI due to their neuroanatomical and immunological features. Our findings suggest that biperiden attenuates early neuronal injury and modulates neurovascular and glial responses. However, because the study was restricted to a single acute time point and did not include direct measures of cholinergic activity or a SHAM-BIP group, these effects should be interpreted as preliminary evidence of potential neuroprotection rather than definitive mechanistic proof.

In our study, the LFPI model was chosen for its extensive validation across different species and its ability to consistently reproduce histological, physiological, metabolic, and behavioral changes that resemble key aspects of human TBI[11]. Notably, the quasi-gyrencephalic organization of the marmoset brain adds translational value and a vital biomechanical dimension to this model. The presence of cortical folding creates variability in force distribution during injury, with the depths of the sulci stress-concentration sites that amplify local strain and increase tissue vulnerability—key characteristics of human TBI that are not present in lissencephalic models[33]. However, the marmoset cortex is not fully gyrencephalic. Its cortical folding is modest compared with the highly folded human brain, and this anatomical difference may influence how the LFPI pulse is biomechanically distributed. Therefore, stress-concentration effects associated with deeper, more intricate sulcal patterns may be underrepresented. Consequently, while LFPI reproduces key aspects of TBI and benefits from application in a partially folded cortex, species-specific differences in cortical architecture should be considered when interpreting regional injury patterns and extrapolating these findings to human TBI.

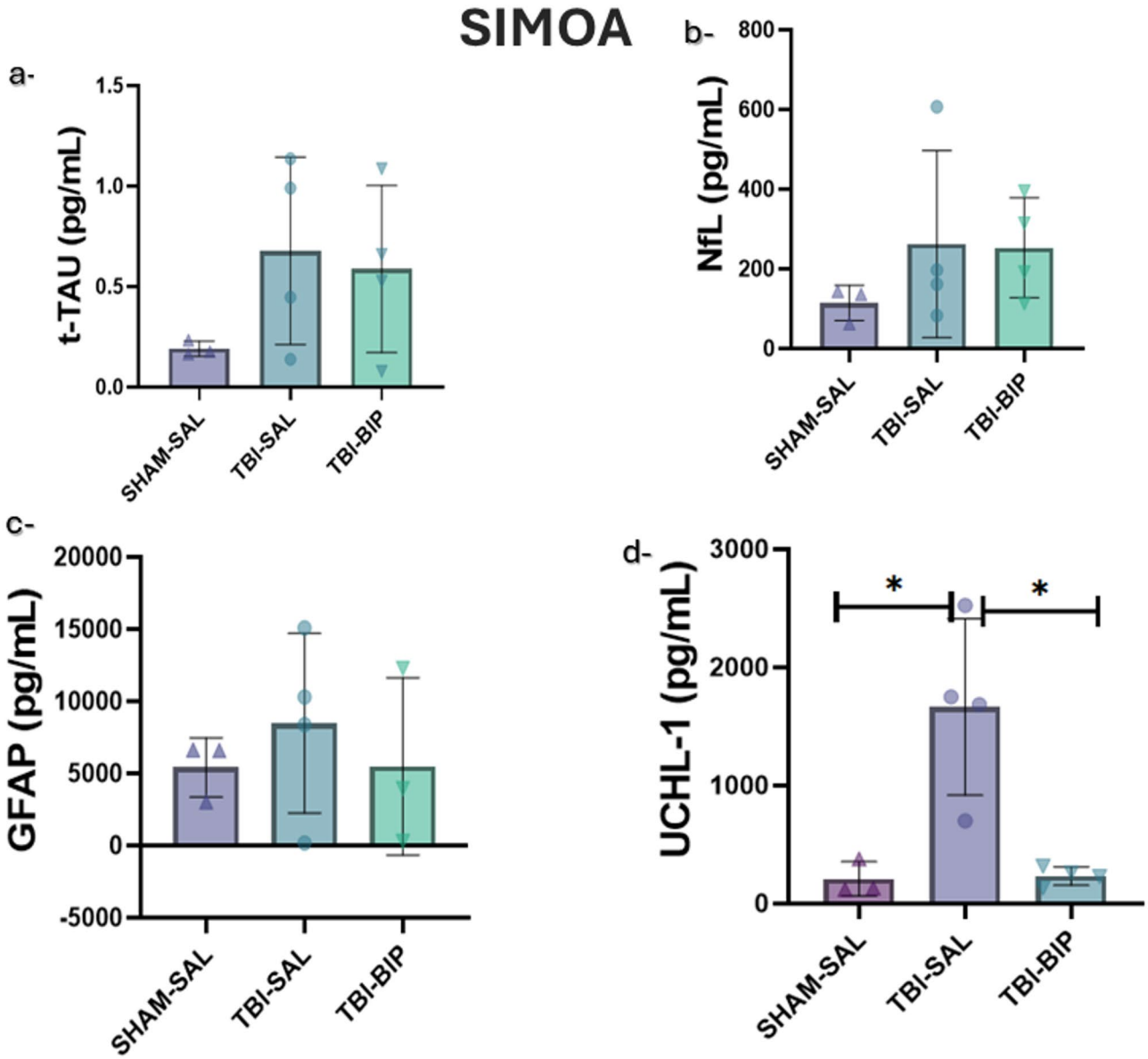


Fig. 7. SIMOA analysis. **a–S**ingle Molecule Array (SIMOA) among the groups **I**-t-Tau, **II**-Neurofilament Light Chain (NfL), **III**- Glial Fibrillary Acidic Protein (GFAP), **IV**- Ubiquitin C-Terminal Hydrolase L1 (UCHL-1). Data are shown as the mean ± standard deviation of the mean. Levels marked with asterisk (*) indicate significant differences: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

At the systemic level, TBI-SAL animals showed increased serum UCH-L1, a marker of neuronal injury and blood–brain barrier (BBB) disruption, whereas this increase was not observed in TBI-BIP animals. These findings support the early neuroprotective effects of biperiden. UCH-L1 is a neuronal protein involved in ubiquitin-mediated protein degradation, and its elevation in serum is commonly associated with neuronal damage and BBB permeability[34]. Accordingly, the increase in the TBI-SAL animals reflects early injury, while its absence in TBI-BIP animals suggests reduced neuronal damage and/or preserved BBB integrity with acute injury, in line with clinical evidence linking UCH-L1 levels to TBI severity group during the acute phase[34,35].

Interestingly, while previous work from our laboratory showed no differences in plasma UCH-L1 levels among groups 24 h after TBI in male rats[32], the present study in marmosets revealed significant increases in serum UCH-L1 levels, consistent with findings from Saletti and collaborators (2023)[36]. In contrast, Morris and colleagues suggested that UCH-L1 may not be a TBI-specific biomarker but rather reflect broader ischemic or neurovascular injury, which aligns with the possibility that its elevation may reflect neuronal stress rather than trauma per se[37].

Consistent with these findings, GFAP and t-Tau showed non-significant increases in TBI-SAL animals, suggesting early astroglial activation and neuronal stress. GFAP is widely associated with astrocytic injury[38],

while t-Tau reflects acute neuronal damage and ongoing neurodegeneration[39]. Together, these trends align with the biochemical profile of early TBI.

Histologically, TBI-SAL animals exhibited larger lesion volumes than the Naive group, with subdural and epidural hematomas present in both TBI conditions. Interestingly, the incidence of hematomas was approximately 25% higher in the TBI-SAL group than in the TBI-BIP animals. However, given that lesion expansion evolves over days[40], the absence of a well-defined cavity at 24 h is expected. Importantly, although the TBI-BIP group was not statistically different from Naive animals in lesion volume, it also did not differ from TBI-SAL animals, indicating that any effect of biperiden on lesion evolution may be modest or not yet detectable at 24 h. Moreover, the absence of statistically significant differences between TBI-BIP and Naive animals should not be interpreted as evidence of complete normalization or treatment efficacy. This highlights the necessity for future longitudinal studies to assess lesion progression and the potential neuroprotective effects of biperiden over time, which could better inform its therapeutic potential.

At the cellular level, TBI-SAL animals exhibited increased neuronal degeneration in the cortex and CA1 regions compared to Naive controls. Moreover, in the CA1, a larger number of blood vessels were observed in the TBI-BIP and SHAM-SAL groups than in the TBI-SAL group, suggesting that untreated TBI animals exhibited reduced vascularization. After TBI, the neurovascular unit is disrupted. Continuous blood flow supplying oxygen and glucose is vital for brain integrity; when flow is reduced or fails to meet metabolic demands, neuronal function declines, and delayed cell death pathways are activated[40,41]. Early alterations in cerebral vasculature are evident after TBI, and these vascular deficits do not appear to fully recover, persisting for weeks to months after the initial injury[42].

In contrast, the TBI-BIP group showed higher TL-positive cell counts in CA1 than the TBI-SAL group, indicating enhanced functional recovery and potentially improved cerebral blood flow. Although the difference between TBI-BIP and TBI-SAL was statistically significant only in the CA1 region, TL-positive vessel counts were consistently higher in TBI-BIP across regions. A similar increase was observed in the TBI-BIP group compared to the NAIVE group in the parietal cortex and CA3 region. Furthermore, the TBI-BIP group displayed a broader distribution of cell density, despite the absence of other significant differences.

Reduced cerebral blood flow contributes to an excitotoxic cascade characterized by excessive glutamate release and accumulation of toxic metabolites in the extracellular space. This process leads to rapid neuronal death accompanied by intense astrocytic activation[40]. Therefore, the lower TL-positive vessel counts observed in our TBI-SAL group may be related to a reduction in GFAP and Iba-1 branch number and average branch length in this group, as well as to the increased number of FJ-B positive cells in the cortex and CA1 of the TBI-SAL group.

Moreover, a study comparing changes in BBB permeability in gyrencephalic and lissencephalic models following TBI suggests that gyrencephalic brain structures may be more vulnerable to vascular disruption than lissencephalic models. Cerebrovascular alterations were observed in ferret brains after blast TBI, including microvascular bleeds followed by significant astrocytosis and microglial activation[43].

GFAP-positive astrocytes in the TBI-BIP group displayed longer branches in the motor cortex, parietal cortex, temporal cortex, CA1, CA3, and DG regions than in the TBI-SAL group, indicating a potentially enhanced capacity for synaptic support and tissue repair. Regarding astrocytes and microglia, our data showed a reduced number of branches in the TBI-SAL and TBI-BIP groups compared to Naive and SHAM-SAL animals 24 h after injury. This reduction may reflect the early activation of resident glial cells, as astrocytes and microglia are among the first to initiate the post-traumatic inflammatory cascade[44,45]. A similar decrease in glial complexity within the ipsilateral cortex has been reported by Villapol and colleagues, who showed hypertrophic astrocytes in the lesion and peri-lesional areas three days post-TBI[40]. Dihné and colleagues (2001) described an acute decrease in astrocyte number and immunoreactivity within 24 h after TBI, followed by a reactive increase through proliferation and hypertrophy, leading to astrogliosis at later stages[46]. Furthermore, astrocytic apoptosis can occur as early as 6 h after trauma, and has been proposed to precede neuronal degeneration by several hours[47].

Following CNS injury, microglia undergo substantial morphological remodeling, characterized by retraction of their processes and adoption of an amoeboid phenotype[48]. Studies have also demonstrated that Iba-1 expression is upregulated in activated microglia compared with their non-reactive, surveillant counterparts[32,48].

Curiously, the hierarchical analysis revealed that, with respect to the number of microglial branches, the SHAM-SAL and Naïve groups separate from the TBI groups, except in CA3 and DG, where the SHAM-SAL group separates from the others. Regarding the number of astrocytic branches, the SHAM-SAL group clustered separately from all other groups, with the greatest number of GFAP+ branches.

This is particularly concerning, as Sham animals demonstrate significant neuronal degeneration in certain regions, increased vessel counts, and distinct astrocytic and microglial clustering patterns. Therefore, SHAM-SAL animals should be interpreted as surgical controls rather than as uninjured baseline controls. This raises important questions about the biological impact of the craniectomy procedure, even though we maintained dura mater integrity during the experiments. One plausible explanation for this distinction is the craniectomy procedure itself, which has been shown in previous studies to cause measurable structural and functional alterations in the underlying brain tissue. This is likely due to the disruption of the intricate network of nerve fibers and blood vessels connecting the brain to the skull during bone flap removal[49]. Craniectomy can also alter intracranial pressure dynamics, increasing the hydrostatic pressure gradient and promoting transcapillary leakage[50]. Although precautions were implemented in our protocol, the use of an electric drill may still generate heat and vibration capable of damaging vascular and neural structures[51]. Moreover, as was mentioned before, gyrencephalic brain structures may be more vulnerable to vascular disruption than lissencephalic models[52].

In contrast, when examining the average branch length of astrocytes, two main clusters emerged: one composed of the Naïve and TBI-BIP groups, and another comprising the remaining groups. This pattern suggests that biperiden may help preserve aspects of glial homeostasis following injury. One possible explanation

is related to the known pharmacological profile of biperiden. After TBI, excessive cholinergic signaling has been associated to neuronal hyperexcitability and glutamate-mediated excitotoxicity[32,52,53]. Because biperiden is pharmacologically characterized as a muscarinic antagonist at M1 muscarinic receptors, muscarinic modulation represents one plausible mechanism by which the drug could influence excitability, as it competes with acetylcholine in a 1:1 manner and modulates receptor activity[54], and may help attenuate early post-traumatic neuronal stress and excitotoxic processes[55,56].

While receptor-level target engagement was not directly assessed in this study, biperiden is well characterized as a preferential M1 muscarinic receptor antagonist[54,57,58] with higher affinity for forebrain regions such as the cortex and hippocampus[58]. Because M1 receptor signaling is closely linked to neuronal excitability, its inhibition may represent a potential mechanism for reducing hyperexcitability and excitotoxicity following SE[24,27] and possibly after TBI as well. However, this interpretation remains indirect and requires further experimental confirmation.

The pronounced astrocytic hypertrophy observed in the TBI-SAL group may reflect a maladaptive glial response, potentially impairing neuronal plasticity, as described by Burda and colleagues[59]. Moreover, our correlation matrix shows a strong correlation between GFAP and Iba-1 branch count, as does the PCA analysis. The PCA analysis indicated that astrocytic and microglial morphological complexity, GFAP, and Iba-1 average branch length vary independently. Suggesting that branching features are the dominant source of variation across samples, and the number of vessels potentially reflects structural remodeling in response to TBI and to biperiden.

Taken together, these findings are consistent with the cholinergic mechanism that contribute to early post-traumatic responses, not only in PTE but also in epileptogenesis. Recent studies have demonstrated that anticholinergic interventions can modulate this process; for example, prolonged scopolamine treatment after lithium–pilocarpine–induced SE markedly reduced spontaneous recurrent seizures (SRS) at six months, even though its early effects were minimal[60]. Consistent with this, work from our group has shown that biperiden administration following pilocarpine-induced SE reduces both the number and the intensity of seizures[24].

While the effects of biperiden provide important insights, they do not fully account for the range of mechanisms underlying acute post-traumatic outcomes, including seizure susceptibility and severity. Other pathways may therefore contribute to these processes. Among them, the mTOR signaling pathway stands out as a key regulator of cellular growth, inflammatory modulation, and neuroplasticity[61]. As a central integrator of mitogenic and nutrient signals, mTOR plays a fundamental role in neuronal survival, axonal and dendritic development, and synaptic plasticity[61–64]. It also regulates protein synthesis in neuronal soma and axons, influencing soma size, growth cone formation, and structural remodeling—processes essential for axonal regeneration. Notably, pharmacological inhibition of mTOR with rapamycin has shown beneficial effects in experimental models of TBI, including reductions in epileptogenesis and aberrant inflammation[61]. Although not directly assessed in the present study, the broad involvement of mTOR in post-injury responses underscores its relevance as a promising target for future investigation beyond the mechanisms associated with biperiden.

This study has several limitations. The analysis was restricted to a single acute time point (24 h), which precludes assessment of subacute and chronic outcomes, including cognitive function. This focus was intentional to isolate early injury mechanisms, such as excitotoxicity, neuronal damage, and glial activation, but it limits conclusions about long-term functional recovery. In addition, no direct measures of cholinergic activity were included, preventing confirmation that the observed effects are specifically mediated by muscarinic receptor antagonism. The effects observed after biperiden treatment are consistent with a potential acute neuroprotective action, possibly related to its muscarinic antagonist profile, although direct target engagement was not assessed in the present primate model.

Further considerations include using animals from a conservation facility, which may introduce variability related to prior exposures and heterogeneity in age and weight. No consistent sex-related differences were observed across experimental groups or brain regions. Although some variability was observed, our study was not powered to formally assess sex as a biological variable, given the low number of animals per sex per group (see Supplemental Material Table S1 and S2). Finally, the absence of a Sham-biperiden group limits the ability to distinguish frug effects on surgical injury from TBI-specific responses. Nevertheless, this work represents an important first step toward understanding the effects of biperiden on marmosets.

In summary, this study provides the first characterization of the acute effects of biperiden following moderate TBI induced by LFPI in a non-human primate model. Untreated TBI animals exhibited significantly elevated UCH-L1 levels, indicating greater neuronal injury and BBB disruption, whereas biperiden-treated animals showed a partial attenuation of this response. Histological analyses suggested trends toward preserved vascular integrity and neuronal viability in hippocampal regions (CA1 and CA3) following biperiden treatment. However, in glia analyses, the biperiden effect was less pronounced, with reduced glial branching observed in both TBI groups; biperiden-treated animals exhibited longer astrocytic and microglia processes, suggesting improved glial homeostasis and repair potential. Overall, these findings suggest that acute biperiden treatment may modulate early neurovascular and inflammatory responses following TBI. However, its neuroprotective efficacy remains limited and should be interpreted with caution, mainly due to the high variability in TBI-BIP. Importantly, the absence of a SHAM-BIP control group, the limited sample size, and the lack of direct cholinergic or molecular pathway validation constrain mechanistic interpretation. Further studies incorporating appropriate controls, longer follow-up periods, behavioral outcomes, and molecular analyses of cholinergic and mTOR-related pathways are required to clarify the therapeutic potential of biperiden in TBI.

Data availability

The datasets used and/or analyzed during the current study are available at the following link https://osf.io/zq5am/overview upon reasonable request.

Received: 30 March 2026; Accepted: 11 May 2026
Published online: 18 June 2026

References

1. Menon, D. K., Schwab, K., Wright, D. W. & Maas A. I. Position statement: Definition of traumatic brain injury. *Arch. Phys. Med. Rehabil.* **91**, 1637–1640 (2010).
2. Brett, B. L., Gardner, R. C., Godbout, J., Dams-O'Connor, K. & Keene, C. D. Traumatic Brain Injury and Risk of Neurodegenerative Disorder. *Biol. Psychiatry*. **91**, 498–507 (2022).
3. Sivanandam, T. M. & Thakur, M. K. Traumatic brain injury: A risk factor for Alzheimer's disease. *Neurosci. Biobehav Rev.* **36**, 1376–1381 (2012).
4. Pitkänen, A. et al. Epileptogenesis in experimental models. *Epilepsia* **48**, 13–20 (2007).
5. Teasdale, G. & Jennett, B. Assessment of coma and impaired consciousness. A practical scale. *Lancet* **37**, 52–56 (1974).
6. Maas, A. I. R. et al. Traumatic brain injury: progress and challenges in prevention, clinical care, and research. *Lancet Neurol.* **21**, 1004–1060 (2022).
7. Xiong, Y., Mahmood, A. & Chopp, M. Animal models of traumatic brain injury. *Nat. Rev. Neurosci.* **14**, 128–142 (2013).
8. McIntosh, T. K. et al. Traumatic brain injury in the rat: characterization of lateral fluid-percussion model. *Neuroscience* **28**, 233–244 (1989).
9. Dixon, C. E. et al. A fluid percussion model of experimental brain injury in the rat. *J. Neurosurg.* **67**, 110–119 (1987).
10. Namjoshi, D. R. et al. Merging pathology with biomechanics using CHIMERA (Closed-Head Impact Model of Engineered Rotational Acceleration): a novel, surgery-free model of traumatic brain injury. *Mol. Neurodegener.* **9**, 55 (2014).
11. Thompson, H. J. et al. Lateral Fluid Percussion Brain Injury: A 15-Year Review and Evaluation. *J. Neurotrauma.* **22**, 42–75 (2005).
12. Cenci, M. A., Whishaw, I. Q. & Schallert, T. Animal models of neurological deficits: How relevant is the rat? *Nat. Rev. Neurosci.* **3**, 574–579 (2002).
13. Vink, R. Large animal models of traumatic brain injury. *J. Neuro Res.* **1–9** https://doi.org/10.1002/jnr.24079 (2017).
14. Finnie, J. W. Animal models of traumatic brain injury: a review. *Aust Vet. J* **79**, 628–633 (2001).
15. Zhang, N. & Zhang, Y. Correlation between gyral size, brain size, and head impact risk across mammalian species. *Brain Res.* **1828**, 148768 (2024).
16. Prins, N. W. et al. Common Marmoset (Callithrix jacchus) as a Primate Model for Behavioral Neuroscience Studies. *J. Neurosci. Methods*. **1**, 35–46 (2017).
17. Shinmyo, Y., Hamabe-Horiike, T., Saito, K. & Kawasaki, H. Investigation of the Mechanisms Underlying the Development and Evolution of the Cerebral Cortex Using Gyrencephalic Ferrets. *Front. Cell. Dev. Biol.* **10**, 1–12 (2022).
18. Donat, C. K. et al. Effects of lateral fluid percussion injury on cholinergic markers in the newborn piglet brain. *Int. J. Dev. Neurosci.* **28**, 31–38 (2010).
19. Homman-Ludiye, J. & Bourne, J. A. The Marmoset: The Next Frontier in Understanding the Development of the Human Brain. *ILAR J.* **61**, 248–259 (2020).
20. McIntosh, T. K. Neurochemical sequelae of traumatic brain injury: therapeutic implications. *Cerebrovasc. Brain Metab. Rev.* **6**, 109–162 (1994).
21. Gorman, L. K., Fu, K., Hovda, D. A., Becker, D. P. & Katayama, Y. Analysis of acetylcholine release following concussive brain injury in the rat. *J. Neurotrauma*. **6**, 203 (1989).
22. Gorman, L., Fu, K., Hovda, D. A., Murray, M. & Traystman, R. J. Effects of Traumatic Brain Injury on the in the Rat. *J. Neurotrauma*. **13**, 457–463 (1996).
23. Andrade, P. et al. Seizure Susceptibility and Sleep Disturbance as Biomarkers of Epileptogenesis after Experimental TBI. *Biomedicines* 10, (2022).
24. Bittencourt, S. et al. Modification of the natural progression of epileptogenesis by means of biperiden in the pilocarpine model of epilepsy. *Epilepsy Res* **138**, 88–97 (2017).
25. Foresti, M. L. et al. Initial clinical evidence on biperiden as antiepileptogenic after traumatic brain injury — a randomized clinical trial. *Front. Neurol.* 1–10. https://doi.org/10.3389/fneur.2024.1443982 (2024).
26. Foresti, M. L. et al. Biperiden for prevention of post-traumatic epilepsy: A protocol of a double-blinded placebo-controlled randomized clinical trial (Biperiden trial). *PLoS One*. **17**, 1–11 (2022).
27. Benassi, S. K. et al. Two decades of research towards a potential first anti-epileptic drug. *Seizure* **90**, 99–109 (2021).
28. Paxinos, G., Watson, C., Petrides, M., Rosa, M. & Tokuno, H. The marmoset brain in stereotaxic coordinates. *Marmoset Brain Stereotaxic Coordinates*. https://doi.org/10.1007/978-0-387-78385-7 (2012).
29. Zeinali, R., Keshtkar, A., Zamani, A. & Gharehaghaji, N. Brain volume estimation enhancement by morphological image processing tools. *J. Biomed. Phys. Eng.* **7**, 379–388 (2017).
30. Boyko, M. et al. Traumatic brain injury-induced submissive behavior in rats: link to depression and anxiety. 1–10 (2022). https://doi.org/10.1038/s41398-022-01991-1
31. Sanabria, V. et al. Standardizing the lateral fluid percussion model of trauma in marmosets (Callithrix jacchus): regional vulnerability and inflammatory response. *Brazilian J. Med. Biol. Res.* **59**, e15289 (2026).
32. Braga, M. et al. Effects of acute biperiden treatment following traumatic brain injury in male rats. *Sci. Rep.* **29**, 44775 (2025).
33. Mazurkiewicz, A. et al. Impact-induced cortical strain concentrations at the sulcal base and its implications for mild traumatic brain injury. *J. Biomech. Eng.* **143**, 1–12 (2021).
34. Papa, L. et al. UCH-L1 is a novel biomarker for severe traumatic brain injury in human. *Crit. Care Med.* **38**, 138–144 (2010).
35. Brophy, G. M. et al. Biokinetic analysis of ubiquitin C-terminal hydrolase-L1 (UCH-L1) in severe traumatic brain injury patient biofluids. *J. Neurotrauma*. **28**, 861–870 (2011).
36. Saletti, P. G. et al. Early preclinical plasma protein biomarkers of brain trauma are influenced by early seizures and levetiracetam. *Epilepsia Open*. **8**, 586–608 (2023).
37. Morris, M. C. et al. UCH-L1 is a Poor Serum Biomarker of Murine Traumatic Brain Injury After Polytrauma. *J. Surg. Res.* **244**, 63–68 (2019).
38. Sacco, M. A. et al. The Role of GFAP in Post-Mortem Analysis of Traumatic Brain Injury: A Systematic Review. *Int. J. Mol. Sci.* **26**, 185 (2025).
39. Schöll, M. et al. Molecular and Cellular Neuroscience Biomarkers for tau pathology. *Mol. Cell. Neurosci.* **97**, 18–33 (2019).
40. Villapol, S., Byrnes, K. R. & Symes, A. J. Temporal dynamics of cerebral blood flow, cortical damage, apoptosis, astrocyte-vasculature interaction and astrogliosis in the pericontusional region after traumatic brain injury. *Front. Neurol.* **5 JUN**, 1–9 (2014).
41. Westergren, H., Ryding, E., Zbornickova, V. & Hillman, J. Traumatic Occlusion of the Internal Carotid Artery in a Healthy Young Male: Effects on the Regional Cerebral Blood Flow. *Acta Neurochir. (Wien)*. **113**, 91–95 (1991).
42. Salehi, A., Zhang, J. H. & Obenaus, A. Response of the cerebral vasculature following traumatic brain injury. *J. Cereb. Blood Flow. Metab.* **37**, 2320–2339 (2017).
43. Rao, K. V. R. et al. Comparison of blood-brain barrier permeability changes in gyrencephalic (ferret & non-human primate) and lissencephalic (rat) models following blast overpressure exposures. *Exp. Neurol.* **392**, 115375 (2025).
44. Hernandez-Ontiveros, D. G. et al. Microglia activation as a biomarker for traumatic brain injury. *Front. Neurol.* **4 MAR**, 1–9 (2013).

45. Diaz-Arrastia, R. et al. Pharmacotherapy of Traumatic Brain Injury: State of the Science and the Road Forward : Report of the Department of Defense Neurotrauma Pharmacology Workgroup. *J. Neurotr*. **158**, 135–158 (2014).
46. Dihné, M., Block, F., Korr, H. & Töpper, R. Time course of glial proliferation and glial apoptosis following excitotoxic CNS injury. *Brain Res.* **902**, 178–189 (2001).
47. Huang, T. et al. Traumatic injury activates MAP kinases in astrocytes: Mechanisms of hypothermia and hyperthermia. *J. Neurotrauma*. **26**, 1535–1545 (2009).
48. Caplan, H. W. et al. Spatiotemporal Distribution of Microglia After Traumatic Brain Injury in Male Mice. *ASN Neuro* **12**, 1759091420911770 (2020).
49. Cole, J. T. et al. Craniotomy: True sham for traumatic brain injury, or a sham of a sham? *J. Neurotrauma*. **28**, 359–369 (2011).
50. Vitali, M. et al. Decompressive Craniectomy in Severe Traumatic Brain Injury: The Intensivist's Point of View. *Diseases* **11**, 1–10 (2023).
51. Santana-Gomez, C. et al. The Surgical Method of Craniectomy Differentially Affects Acute Seizures, Brain Deformation, and Behavior in a Traumatic Brain Injury Animal Model. *Neurotrauma Rep.* **5**, 969–981 (2024).
52. Shin, S. S. & Dixon, C. E. Alterations in cholinergic pathways and therapeutic strategies targeting cholinergic system after traumatic brain injury. *J. Neurotrauma*. **32**, 1429–1440 (2015).
53. Dixon, C. E. et al. Time course of increased vulnerability of cholinergic neurotransmission following traumatic brain injury in the rat. *Behav. Brain Res.* **70**, 125–131 (1995).
54. Hamilton, S. E. et al. Alteration of cardiovascular and neuronal function in M 1 knockout mice. *Life Sci.* **68**, 2489–2493 (2001).
55. Sanabria, V. et al. Anticholinergics: A potential option for preventing posttraumatic epilepsy. *Front. Neurosci.* 1–8. https://doi.org/10.3389/fnins.2022.1100256 (2023).
56. Scarr, E. Muscarinic Receptors: Their Roles in Disorders of the Central Nervous System and Potential as Therapeutic Targets. *CNS Neurosci. Ther.* **18**, 369–379 (2012).
57. Burke, R. E. The relative selectivity of anticholinergic drugs for the M1 and M2 muscarinic receptor subtypes. *Mov. Disord*. **1**, 135–144 (1986).
58. Eltze, M. & Figala, V. Affinity and selectivity of biperiden enantiomers for muscarinic receptor subtypes. *Eur. J. Pharmacol.* **158**, 11–19 (1988).
59. Burda, J. E., Bernstein, A. M., Sofroniew, M. V. & Angeles, L. Astrocyte roles in traumatic brain injury. *Exp. Neurol.* **275**, 305–315 (2016).
60. Meller, S., Käufer, C., Gailus, B., Brandt, C. & Löscher, W. Scopolamine prevents aberrant mossy fiber sprouting and facilitates remission of epilepsy after brain injury. *Neurobiol Dis* **158**,105446 (2021).
61. Don, A. S. A. et al. Targeting mTOR as a novel therapeutic strategy for traumatic CNS injuries. *Drug Discov Today*. **17**, 861–868 (2012).
62. Fan, J. et al. PDX1 phosphorylation at S61 by mTORC1 links nutrient signaling to β cell function and metabolic disease. *Cell. Rep.* **45**, 1–37 (2026).
63. Zhao, T., Fan, J., Abu-Zaid, A., Burley, S. K. & Zheng, X. F. S. Nuclear mTOR Signaling Orchestrates Transcriptional Programs Underlying Cellular Growth and Metabolism. *Cells* **13**, 781 (2024).
64. Fan, J., Yuan, Z., Burley, S. K., Libutti, S. K. & Zheng, X. F. S. Amino Acids Control Blood Glucose Levels through mTOR Signaling. *Eur. J. Cell. Biol.* **101**, 1–7 (2022).

Acknowledgements

In memory of our dear veterinarian, Paulo Varoni Cavalcanti. We also extend our sincere thanks to Professor Fábio Cruz for kindly allowing us to use his microscope, to Master Simone Cinini for her help with the marmoset's transportation, and to Mery Liz Alfaro for her help.

Author contributions

V. S.: Conceptualization, Methodology, Formal Analysis, Validation, and Data Curation, Writing- Original Draft. C. G.: Conceptualization, Methodology, Formal Analysis, Review. S. R.: Methodology and Review. (A) S. G.: Methodology and Review. M. (B) B.: Methodology and Review. M. L. C.: Methodology, Formal Analysis, Validation, and Data Curation, Review. J. L. W.: Methodology and Review. M. L. F. : Conceptualization, Validation, Review, and Editing. L. E. M.: Conceptualization, Review, and Editing. B. M. L.: Conceptualization, Methodology, Review, Editing, and Supervision.

Funding

Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP grants 2018/24561-5 and 2022/00249-8), Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil (CAPES; Finance Code 001), and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq; grants 311619/2019-3, and 312904/2021-5) supported this work. VS and SR (2017/05242-3) received post-doctoral fellowships from the São Paulo Research Foundation (FAPESP).

Declarations

Competing interests

The authors declare no competing interests.

Consent for publication

All authors have agreed to the published version of this manuscript.

Ethics statement

All procedures were conducted in compliance with ARRIVE, NIH, and IBAMA (Brazilian Institute of Environment and Renewable Natural Resources) guidelines, and were approved by the institutional Animal Care and Use Committee (protocol #5138271222).

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-53148-5.

Correspondence and requests for materials should be addressed to V.S. or B.M.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026